

Partial Derivatives

Benjamin Abshire[†]

February 10, 2026

I Introduction

Partial derivatives allow us to study how a function changes with respect to one variable, holding all others constant. Formally, the partial derivative of $f(x, y)$ exists if

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0 - y_0)}{h}$$

exists. We denote partial derivatives by

$$\frac{\partial f}{\partial y}(x_0, y_0) = f_y(x_0, y_0)$$

In the general case, if $f(x_1, x_2, \dots, x_n)$ then

$$\frac{\partial f}{\partial x_j} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_j, x_{j+h}, \dots, x_n) - f(x_j - x_n)}{h}$$

We regard all variables besides the ones we are differentiating with respect to as constants. For example, if we are differentiating $f(x, y)$ with respect to x , we consider y a constant.

II Examples

Example 1. $f(x, y) = 2x^3 + y^2$

Solution.

$$\frac{\partial f}{\partial x} = 6x^2, \quad \frac{\partial f}{\partial y} = 2y$$

Example 2. $f(x, y) = x^2 + 2xy$

Solution.

$$\frac{\partial f}{\partial x} = 2x + 2y, \quad \frac{\partial f}{\partial y} = 2x$$

[†]Definitions and example problems from MATH 22005: Analytic Geometry and Calculus III instructed by Dr. Gang Yu. Kent State University.

Example 3. $f(x, y) = (x^2 + 2xy)e^{x+y}$

Solution. Here we just apply the product and chain rules, and pay attention to which variable is considered a constant.

$$\frac{\partial f}{\partial x} = (2x + 2y)e^{x+y} + (x^2 + 2xy)e^{x+y} \cdot 1,$$

$$\frac{\partial f}{\partial y} = (2x)e^{x+y} + (x^2 + 2xy)e^{x+y} \cdot 1$$

Example 4. $f(x, y) = \cos(xy)$

Solution. Here we just apply the chain rule, and pay attention to which variable is considered a constant.

$$\frac{\partial f}{\partial x} = -\sin(xy) \cdot y, \quad \frac{\partial f}{\partial y} = -\sin(xy) \cdot x$$

III Partial Derivatives of Higher Order

If we want to take the second derivative of $f(x, y)$ with respect to x we can write

$$\frac{\partial^2 f}{\partial x^2} = f_{xx}(x, y) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$$

If we want to take the derivative of $f(x, y)$ with respect to x then with respect to y , we can write

$$\frac{\partial^2 f}{\partial x \partial y} = f_{xy}(x, y) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

Note that

$$f_{xy}(x, y) = f_{yx}(x, y)$$

Example 5. $f(x, y) = x^2 + 2xy$

Solution.

$$f_{xx} = 2, \quad f_{xy} = 2, \quad f_{yx} = 2$$